Kai Sprunger

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Education

University of Central Florida, Burnett Honors College — Orlando, FL
Bachelor of Science in Computer Science, GPA: 3.7 — *Expected graduation: May 2028*

Experience

Software Engineer Intern — May 2026 – Aug 2026
NVIDIA — *Santa Clara, CA*
- Made **agentic AI Python pipeline** to develop MISRA/CERT violation fixes and write deviations for **70+ files**.
- Made **Jenkins** pipeline to flag MISRA/CERT violations and post suggestions on **GitLab** MRs.
- Fixed **600 violations**, authored **900 deviations**, cutting MISRA/CERT-C findings from **1,800 to 200**, an **89% drop**.
- Built **Python** script to scan for **Coverity** violations, apply profile and ruleset configs, and produce **HTML/CSS** reports.

Software Engineer Intern — Jan 2026 – Apr 2026
Bank of New York Mellon — *Orlando, Florida*
- Made **4-agent** observability pipeline for **AppDynamics, Grafana, Prometheus** that resolved **500+ injection attacks.**
- Trained **Isolation Forest model** on **4,000+ points** over **8 features** and **300+ trees** to flag **latency** and **Docker** faults.
- Built **Random Forest + XGBoost model** to classify **6 fault causes** with **85% accuracy**, automating **70%** of paths.
- Integrated **Google Gemini** as a fallback, querying incident and anomaly context from a **PostgreSQL database.**

Software Engineer Intern — May 2025 – Aug 2025
NVIDIA — *Santa Clara, CA*
- Wrote **10,000+ C microkernel tests** across **230 RISC-V OS functions**, raising line coverage from **77.7% to 97.1%**.
- Shipped **4 features** for **8 critical inter-process communication and thread-local storage bugs** to production.

Software Engineer — Aug 2024 – Apr 2026
IEEE @ UCF — *Orlando, FL*
- Led **25-developer team** for **4 products**, boosting output by **600%** via DevOps, GitHub workflows, and ticketing.
- Developed **6 React/Tailwind pages** and **35 frontend components**, deploying the **full-stack website** on **Vercel**.
- Deployed **PostgreSQL/Drizzle ORM database** and built **Discord OAuth authentication system** for **100+ users**.
- Made **39 tRPC backend endpoints** over **7 routers**, backing **CRUD** for **100+ users, 200+ events, and 6 projects**.

Projects

Agentic Nuclear Reactor (HackUSF 2026 Winner) | *Google ADK, Gemini, FastAPI, WebSocket, Arduino* — Mar 2026
- **1st Overall & Google Cloud Winner from 80+ teams**: agentic pipeline regulating a reactor's electrostatic fields.
- Implemented a **15-rule Gemini decision tree** with **Python safety overrides** and meta-agent tuning of operating limits.
- Made an **analysis agent** to detect beam instability through **EWMA, COV, integral error, and rate-of-change**.
- Integrated **Python serial control** with an **Arduino** for MCP4728 DAC voltage output and **WebSocket** telemetry.

SignHero (SwampHacks 2026 Winner) | *React, TypeScript, WebSockets, tRPC, MongoDB* — Jan 2026
- **2nd Place Overall & Best Game Design Winner from 110+ teams**: ASL rhythm game using machine learning.
- Built type-safe **tRPC, Prisma, and MongoDB services** managing player creation, validation, results for **50+ players**.
- Made mistake tracker with **React hooks, Tailwind, WebSocket**, effectively showing missed signs for **150+ rounds.**
- Used **tRPC routes, webhooks, Prisma, MongoDB** for beatmap creation and used **React, Tailwind** for song carousel.

CappuConnect (STEM Connect 2025 Winner) | *MongoDB, Next.js, React, TypeScript, Puppeteer* — Sept 2025
- **Winner of Best Non-AI Hack from 30+ teams**: Networking app with vector-based event suggestions.
- Built **MongoDB REST APIs** for member/event CRUD and integrated NextAuth authentication for **160+ accounts**.
- Implemented **TypeScript vector-similarity ranking** for **600+ scraped events**, personalizing recommendations.

Tariffix (Bitcamp 2025 Winner) | *MongoDB, Cheerio, Gemini, React, Tailwind CSS, Next, Vite* — April 2025
- **MLH Best Use of MongoDB winner from 35+ teams**: Extension/API revealing how tariffs affect product prices.
- Deployed **tariff rate API on MongoDB** and developed **REST API routes with Express and Mongoose**.
- Built extension with **Cheerio and Gemini** to analyze Amazon products, map HTS codes, generate tariff estimates.

Plantopia | *Kotlin, Android SDK, Retrofit/OkHttp, Firebase, Room* — Sept 2025
- Integrated **REST API** with **Retrofit/OkHttp** to retrieve species data and provide watering, sunlight, soil instructions.
- Implemented **Firebase Authentication** and **Room (SQLite) database** to secure user accounts across sessions.

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C, C++, C#, SQL, Verilog, Assembly (RISC-V), Kotlin, Swift, Rust, Go
Frameworks: React, FastAPI, Spring Boot, Next.js, Node.js, Express.js, Tailwind, tRPC, Google ADK, WebSocket, Vue
Libraries: Retrofit, OkHttp, Apache Spark, scikit-learn, NumPy, Pandas, PyTorch, OpenCV, Selenium, Playwright, Drizzle, Prisma, Mongoose, Matplotlib, XGBoost, Puppeteer, NextAuth, pyserial, Redux, Django
Tools: Firebase Authentication, MongoDB, PostgreSQL, SQLite, Neon, Git, Visual Studio Code, Perforce, Docker, Vercel, Figma, Jira, Unity, Jupyter Notebook, Grafana, Prometheus, Arduino, Raspberry Pi, Godot, AppDynamics, Jenkins, Supabase, XCode, iOS, Android SDK